

McMaster Research Ethics Board

Choosing the Application Type

1.2 What kind of research ethics application do you wish to create? Click on the info button for an explanation of each application type.

Note: The choices below are for new projects. If you are looking to make an amendment to an existing study then you need to select the already approved study from your work area and use the "create sub-form" action.

Standard MREB Application

1.3 Is this a post-doc, graduate or undergraduate student project? If so, the Faculty Supervisor's contact information will be requested and the Faculty Supervisor must sign the application prior to submission.

If the project supports both faculty and student research, then select NO, so the faculty researcher can be listed as the Principal Investigator.

- ☐ Yes
- ☒ No

If you are an Instructor for a course-based application, please answer NO to this question. For the purposes of this form, as an Instructor, you will be considered to be a Principal Investigator.

Summary of Revisions in Response to MREB Comments

When you respond to the MREB ethics review sent by the MREB Chair, return to this question and select that option. This will allow you to upload your summary of revisions. If this is the initial submission of your application or you are submitting a revised version due to administrative comments from the Research Ethics Officer, then select that option instead.

1.4 Please select the status of your application:

- ☐ My revised application in response to ethics review comments from the MREB Chair
- ☒ My new application OR my revised application in response to the Research Ethics Officer's administrative comments

Principal Investigator

2.5 Principal Investigator

Title	First Name	Surname
Dr.	Spencer	Smith
Organisation	McMaster University	
City	Hamilton	
Telephone	9055259140 X27929	
Email	smiths@mcmaster.ca	

2.6 What is the faculty/department of the Principal Investigator?

Note for email fields: All email fields are now required to be in **email format**. If you do not want all research team members (Co-investigators, Collaborators and Students) to receive the MacREM email notifications , please enter the following placeholder email instead: applicant@mcmaster.ca.

Note for Title fields: The Title field does not require traditional titles. For example, enter PhD student, Undergrad, Mx, etc.

2.7 Are there any Co-Investigators?

- ☒ Yes
☐ No

2.8 Are there any Collaborators?

- ☐ Yes
☒ No

2.9 Are there any Research Assistants or Coordinators?

- ☐ Yes
☒ No

2.10 Are there any Student Investigators?

- ☒ Yes
- ☐ No

Co-Investigator(s)

2.11 Co-Investigator

Title	First Name	Surname
Dr.	Jacques	Carette
Organisation	McMaster University	
Department	Computing and Software	
Faculty	Engineering	
City	Hamilton	
Telephone	9055259140	
Email	carette@mcmaster.ca	

2.11 Co-Investigator

Title	First Name	Surname
Dr.	Matthew	Giamou
Organisation	McMaster	
Department	Computing and Software	
Faculty	Engineering	
City	Hamilton	
Telephone	905-525-9140	
Email	giamoum@mcmaster.ca	

2.11 Co-Investigator

Title	First Name	Surname
Dr.	Lingyang	Chu
Organisation	McMaster	
Department	Computing and Software	
Faculty	Engineering	
City	Hamilton	
Telephone	905-525-9140	
Email	chul9@mcmaster.ca	

Student Investigator(s)

2.14 Student Investigator(s)

Title	First Name	Surname
Mr	Ziyang	Fang
Organisation	McMaster University	
Department	Computing and Software	
Faculty	Engineering	
City	Hamilton	
Telephone	905-525-9140	
Email	fangz58@mcmaster.ca	

Additional Information

2.15 Additional information about the research team, if applicable (e.g., students or research assistants to be brought on later, this study includes co-supervisors or co-PIs who can both sign the ethics forms, etc.)

Project Title

MREB#: 8215 Title: State of the Practice for Scientific Software

3.1 Provide the full title for your research project:

The State of the Practice for Research Software

3.2 Short Project Title

If your project title is long (more than 15 words), then consider providing a short title below. Otherwise, enter the full title of the project. The title entered below will appear on the project lists in MacREM and a shorter title can save "screen real estate". Please do not use an acronym for the short title.

State of the Practice for Scientific Software

Note: If you are completing this application as a Program of Research for a planned series of studies/experiments in your research lab, then add "- Program of Research" to the end of your full title and short title in 3.1 and 3.2 above. The help button to the right contains some guidance on completing the application as a program of research. Please ensure that the ethics office has confirmed a program of research format will work for your series of studies/experiments before completing the application in this way.

Level of Project

3.3 For which Level of Project(s) will the data be used? (Check all that may apply)

- ☐ Faculty Research
- ☐ Post-Doctoral Research
- ☐ Ph.D. Thesis
- ☒ Master's (Major Research Paper - MRP)
- ☒ Master's (Thesis)
- ☐ Graduate Course Project
- ☐ Staff/Administration Research
- ☐ Undergraduate Honour's Thesis
- ☐ Undergraduate (Independent Research)
- ☐ Doctor of Practical Theology
- ☐ Other

Lay Summary of the Proposed Research

3.4 How would you describe the research activity conducted in this protocol to a lay person unfamiliar with your discipline's methodologies and jargon? (max 250 words)?

Research software (RS) consists of computer programs used by scientists and engineers to investigate unknown areas via simulation and numerical analysis of mathematical models of physical systems. Examples of RS include structural analysis software, machine learning, robot path planning, computational medicine, etc. RS can play an important role for decisions impacting health, safety finances and the environment. Given the importance of RS, scientists and engineers are pushing to improve software qualities such as trustworthiness and reproducibility. Moreover, since RS often has a long life, the development practices need to be sustainable, even when limited developer resources are available. The simultaneous goals of trustworthiness, reproducibility and sustainability are not always achieved today because the current state-of-the-practice for RS does not incorporate state-of-the-art software engineering tools and methods. To have a hope of improving this situation, we need to fully understand the current state of the practice for RS. To understand this and to assist in developing methods for measuring qualities, this project will identify several domains within RS and endeavour to summarize the current state of the practice within those domains. We will look for what is working well in each domain and where there is room for improvement. This "state of the practice" exercise will build off of prior work on measuring/assessing the state of RS software development practice. With this re-boot we will identify a list of software packages within each RS domain and assess their adoption of best practices. We will use quantitative and qualitative analysis for this. Measurements will be based on existing open-source software repositories and interviews with the software developers.

Start and End Dates

3.5 What is the date you plan to begin recruiting participants? Or, for secondary use of data, what is the date you plan to receive the dataset?

If your project has multiple data collection procedures/phases, then provide the date for the earliest planned recruitment.

Note that you cannot begin recruitment or data collection until you have received MREB ethics clearance.

Please note that estimated time from a new full submission (after administrative review) to first response from the REB Chair is approximately 4 - 6 weeks.

Jun/01/2026

3.6 What is the estimated last date for data collection with human participants or, in the case of secondary data, when you intend to complete analysis for this project?

Dec/18/2026

Funding and Granting Agencies

4.1 Is this project currently being funded?

- ☐ Yes
☒ No

4.5 Is funding or additional funding being sought?

- ☐ Yes
- ☒ No

4.7 Are you requesting ethics clearance for a research project that was not originally designed to collect data from human participants or their records (i.e., your research project originally did not involve collecting data from humans or their records but you now intend to do so)?

- ☐ Yes
- ☒ No

Location of the Research

5.1 Select the location(s) where research will be conducted.

- ☒ McMaster University
- ☐ Community
- ☐ Hospital
- ☐ Outside Canada
- ☐ School Boards
- ☒ Online
- ☐ Other

Review by Other Research Ethics Boards

5.2 Has any other Research Ethics Board(s) or equivalent already cleared this project?

- ☐ Yes
- ☒ No

5.4 Will any other Research Ethics Board(s) or equivalent be asked for clearance?

- ☐ Yes
- ☒ No

5.5 Has a version of this study been disapproved or rejected by any Research Ethics Board/Committee?

- ☐ Yes
- ☒ No

Research Involving First Nations, Inuit and Métis Peoples of Canada

6.1 Will your research involve collecting data about First Nations, Inuit or Métis community(ies), include Indigenous identity as a recruitment criteria, or include analysis and interpretation referring to Indigenous communities, people, language, history or culture?

- ☐ Yes
- ☒ No

If you answer No, but are not sure, please answer Yes to see the criteria statements, or contact the Research Ethics Office for more information.

Conflicts of Interest

7.1 Are there any of the following relationships between the investigator(s) and participant(s)? Select all that might apply:

- ☐ Instructor/Teaching Assistant Researcher - Student Participants
- ☐ Manager/Supervisor Researcher - Employee Participants
- ☐ Family Member
- ☐ Friend(s)
- ☐ Student Peers
- ☐ Fellow Club Members
- ☐ Business/Work Colleagues
- ☐ Service Provider Researcher - Client Participants
- ☐ Religious Leader Researcher - Religious Community Participants
- ☒ No Relationship
- ☐ Other

7.4 Will the researcher(s), members of the research team, and/or their partners or immediate family members receive any personal benefits (for example a financial benefit such as remuneration, intellectual property rights, rights of employment, consultancies, board membership, share ownership, stock options etc.) as a result of or being connected to this study?

- ☐ Yes
- ☒ No

7.6 Are there any restrictions regarding access to or disclosure of information (during or at the end of the study) that have been placed on the investigator(s)? These restrictions could come from a study sponsor, community partners, an organization being researched, or another group involved in the study.

- ☐ Yes
- ☒ No

Description of the Project

MREB#: 8215 **Title:** State of the Practice for Scientific Software

8.1 Describe concisely and in plain language the study background of the proposed research project (e.g., context of the research, previous research, importance of this area of study, etc.).

The context of this research is to assess the state of the practice for developing research software. Research software is critical for fields such as nuclear safety analysis, computational medicine and machine learning. Although improving, the current software development approach leads to concerns around trustworthiness, reproducibility and sustainability. Unfortunately, research software project failures have occurred and continue to occur. The complexity and volatility of software has manifested in difficulty in reproducing and replicating scientific results, leading to several retractions. Issues of maintenance and productivity effect research software development. As the software evolves, maintenance cost increase, especially if there are no adequate software engineering practices to support maintenance activities.

The proposed study will update previous research by the applicant looking at the state of the practice of research software. Around 2015 the applicant looked at the state of the practice for several domains of research software, including geographic information systems, mesh generation software and statistics software for psychology. At that time, the research was only based on publicly available data from software repositories. Around 2022 this research was updated with new domains (lattice Boltzmann solvers, medical imaging) and the addition of interviews with developers. This second assessment exercise received ethics approval from MREB under the title "AIMSS-State of the Practice", project number 5219.

8.2 Describe concisely and in plain language the specific purpose / research question for the proposed study.

To understand the attitudes and actions of research software developers towards software quality?

Participants

If researching several sub-populations or groups, use a heading for each population and provide details for each.

9.1 Provide the following information about the participants.

If there are multiple groups/types of participants then list each separately with the requested information.

9.1 Description of the participant group (e.g., McMaster students in Social Science, general population, employees of company XYZ, a) hospital nurses in Cairo, etc.).

Open source research software developers.

9.1 The approximate number of participants for each participant group required for the study. If the study has multiple data collection b) methods, then clarify the number of participants for each method (e.g., 200 participants for the survey and 25 interview participants).

About 60 participants will be invited. Based on the last time we measured the state of the practice, we expect about 10% to 25% of those invited to participate; therefore, we expect 6 to 16 participants. We will be studying two SCS domains: Neural Network libraries, and Robot Motion Planning and Simulation. For each domain we plan to analyze about 30 software packages. The domain of research software is identified using several criteria including: well-defined and stable theoretical underpinning, community size, and open source. The list of candidate software in the domain is found using online or publication based authoritative lists that fall within the identified research software domain. We then filter the list down to a manageable size (30) using several criteria including: functionality, availability of empirical measures, source code format, completeness. We will try to interview one developer of each of these packages.

- 9.1 Any inclusion and/or exclusion criteria for participating in the study (e.g., age, gender, location, health status, immigration status, c) language skills, years of service at an organization, etc.).

None. These will be developers of the software packages of interest.

Language Translation for Participants

- 9.2 Will the study include participants who will need translation of documents or verbal communication from English to another language(s) and/or from another language(s) to English?

This includes both research where some individual participants may need translation and research where it is expected that the entire study population will need the documents presented to MREB in English translated to the local language(s), such as the Letter of Information.

- ☐ Yes
☒ No

Recruitment

If researching several sub-populations, use a heading for each study population and provide details for the type of participant or group and who does the recruiting.

- 10.1 How will each type of participant be recruited?

If researching several sub-populations, use a heading for each study population and provide details for the type of participant or group and who does the recruiting.

The developers of the software packages of interest will be contacted via email using the contact address that is listed on the webpages of their software packages.

- 10.2 Who will recruit each type of participant?

Student investigator.

- 10.3 Will you require permission from an organization or individual in order to proceed with any part of the above recruitment strategies? (e.g., permission from a company to recruit employees on site).

- ☐ Yes
☒ No

Participant Recruiting Methods

MREB#: 8215 Title: State of the Practice for Scientific Software

10.6 Identify the documents that will be used during recruitment (select all that apply):

There are sample recruitment documents in the [Help - Templates](#) section in the menu above. You can sort the documents alphabetically by clicking on Document Name.

- ☐ Recruitment brochure
- ☐ Recruitment poster
- ☐ Video/audio recording
- ☐ Social media (e.g. Instagram, Twitter, etc.)
- ☐ Website Post or Online Advertisement
- ☐ Sona ad (include Debriefing Sheet if applicable - e.g., when using the PNB participant pool)
- ☐ MIRA VOICE platform
- ☐ Crowd sourcing platform (e.g. MTurk, Prolific, etc.)
- ☐ Telephone Verbal Recruitment Script
- ☐ In-person Recruitment Script
- ☐ Hardcopy Letter – e.g. sent via mail
- ☒ Email script/Direct Message (sent direct to participant) and reminder emails (if applicable)
- ☐ Email script (sent from holder of participant contact) and reminder emails (if applicable)
- ☐ Recruitment for follow up interview/focus group
- ☐ Snowball recruitment script
- ☒ Appreciation letter/certificate/thank you card
- ☐ Not Applicable (e.g. secondary use of data, self study)
- ☐ Other

10.6.7 Upload your email script(s) sent directly to participants and reminder emails.

Upload in PDF format

The document file name should clearly indicate type of document, as the file name will appear on the clearance certificate.

If uploading a revised document,

- 1) use track changes so the revisions can be easily verified
- 2) delete the previous version of the document from this upload question.

Documents

Type	Document Name	File Name	Version Date	Version	Size
Recruiting Materials	RecruitingScript	RecruitingScript.pdf	May/11/2026	1	38.4 KB
Recruiting Materials	RecruitingScript_FollowUpEmail	RecruitingScript_FollowUpEmail.pdf	May/11/2026	1	39.3 KB

10.6.12 Upload your appreciation letter/certificate thank you card for participants.

Upload in PDF format

The document file name should clearly indicate type of document, as the file name will appear on the clearance certificate.

If uploading a revised document,

- 1) use track changes so the revisions can be easily verified
- 2) delete the previous version of the document from this upload question.

Documents

Type	Document Name	File Name	Version Date	Version	Size
Recruiting Materials	ThankYouScript	ThankYouScript.pdf	May/11/2026	1	29.2 KB

10.7 Will potential participants answer screening questions to determine eligibility to participate in the study?

- ☐ Yes
- ☒ No

Research Methods

11.1 Describe sequentially, and in detail all data collection procedures in which the research participants will be involved (e.g., paper and pencil tasks, interviews, focus groups, lab experiments, participant observation, surveys, physical assessments etc. —this is not an exhaustive list).

Include information about who will conduct the research (include tasks done by assistants, translators, transcriptionists etc.), how long it will take, where data collection will take place, and the ways in which data will be collected (e.g., computer responses, handwritten notes, audio/video/photo recordings etc.).

The interviewer (a Student Investigator, or Principal Investigator, or Co-Investigator) will conduct a video-conference interview with each interviewee using McMaster's Microsoft Teams. Around 20 technical questions will be asked. The answers will be transcribed into text using Microsoft Teams. The process should take 60 minutes per interview.

11.2 Describe your data analysis methods, (e.g. statistical analysis, textual analysis, NVIVO, analysis resulting in creative outputs, etc.)?

Textual/content analysis. We will be looking for patterns between the answers of all interviewees. We will be looking for a correlation with independently collected data on each project such as the type of documentation provided for the project, the number of lines of code, etc.

Online Surveys and Experiments

11.3 Are you using an online survey or data collection tool?

- ☐ Yes
- ☒ No

Community Based Research

11.5 Are you doing community based research? Click on the info icon to the right for a definition of community based research.

Note: If the research is with an Indigenous community and the community engagement information has already been provided in Section 6, it is not necessary to complete this section and 11.6 should be left as “no”.

- ☐ Yes
- ☒ No

Test Instruments and Interview Guides

11.13 Select and upload copies of all questionnaires, interview guides (i.e., lists of questions), tests, or data collection instruments, etc.

- ☐ Demographic Form
- ☐ Instructions for Participants (e.g. how to perform a task in an experiment, instructions for a Photovoice project, etc.)
- ☒ Interview Guide (face to face, telephone, Skype, etc.)
- ☐ Focus Group Guide (questions for focus group)
- ☐ Sharing Circle or Talking Circle Guide
- ☐ Questionnaire or Survey
- ☐ Rating scales/inventories/assessment instruments
- ☐ Role-play simulation scripts
- ☐ Stimuli used to elicit responses
- ☐ Pictures (or diagrams) of what the participant will experience in the study, such as wearing equipment (e.g. EEG) or doing physical tasks
- ☐ LIVELab SOPs
- ☐ Not Applicable (e.g. secondary use of data, self study)
- ☐ Other

11.13.5 Upload your Interview Guide - (Questions for face to face, telephone, internet/email interview).

Upload in PDF format

The document file name should clearly indicate type of document, as the file name will appear on the clearance certificate.

If uploading a revised document,

- 1) use track changes so the revisions can be easily verified
- 2) delete the previous version of the document from this upload question.

Documents

Type	Document Name	File Name	Version Date	Version	Size
Interviews	InterviewGuide	InterviewGuide.pdf	May/11/2026	1	145.6 KB

Secondary Use of Data

12.1 In this current research project are you planning to use secondary data that was originally collected for another purpose?

- ☐ Yes
- ☐ No

Research Database

12.32 Does your research involve the creation and/or modification of a research database (databank) containing human participant information? A research database is a collection of data maintained for use in future research. The human participant information stored in the research database can be identifiable or anonymous.

- ☐ Yes
- ☒ No

A research database is a formal collection of data maintained and administered by you for use in future research by you and other researchers. This is different from simple data storage of your research data for your own future use. Big Data or databank projects most likely would need to select Yes but most researchers collecting data can select No. Contact the ethics office if not sure.

Incentives

This section asks different questions for incentives, reimbursement and compensation as each are considered different forms of payment.

13.1 Will participants receive an incentive for participation?

- ☐ Yes
- ☒ No

Reimbursements

13.5 Will participants be reimbursed for expenses related to participating in the research (e.g., transportation, parking, childcare, taking unpaid leave from work)?

- ☐ Yes
- ☒ No

Compensation

The application section of [Article 3.2](#) in the TCPS2 notes participants should be informed about any compensation they may be entitled to for research-related injuries. This is only applicable if your study has a genuine likelihood of causing physical injury or financial harm (e.g. job loss) to participants. This will not apply for most studies reviewed by MREB.

13.8 Will participants be entitled to any compensation for research-related injuries?

- ☐ Yes
- ☒ No

Risks and Benefits Inherent in the Research

Indicate if the participants might experience any of the following risks:

Physical Risks

14.1 Physical risks (including any bodily contact or administration of any substance)?

- ☐ Yes
☒ No

Psychological Risks

14.3 Psychological risks (including feeling demeaned, embarrassed, worried or upset)?

- ☒ Yes
☐ No

14.4 Provide the following;

- a) Description of the potential psychological risk(s).
- b) Explanation of how the psychological risk(s) will be managed or minimized.
- c) If the study includes significant psychological risk, then explain why alternative approaches with less psychological risk cannot be used.

- a) There is very little or no risk associated with our study. The interviewee might feel that their work is being scrutinized.

b) We will assure the interviewees that our interview questions are objective and technical. The questions will not suggest that a specific course of (technical) action is better than another. Our questions focus on what was done and not why it was done.

c) The potential psychological risk is very minimal, and there are no alternatives with lesser risks.

Social Risks

14.5 Social risks (including possible loss of status, privacy and / or reputation as well as economic risks)?

- ☒ Yes
☐ No

14.6 Provide the following;

- a) Description of the potential social risk(s).
- b) Explanation of how the social risk(s) will be managed or minimized.
- c) If the study includes significant social risk, then explain why alternative approaches with less social risk cannot be used.

a) Social risk is minimal. The participants face the social risk of a privacy breach if someone deduces their identity and sees their interview responses. We will identify by name the software packages in our research. There is also a risk of data breach by virtue of the interview being conducted online and the interview answers being stored in an online repository.

b) Privacy breach due to the deduction of an identity will be minimized in the following way: While we list the software packages in our research, interviewee names will never be shared. Interviewee names will be secured and confidential, as detailed in Section 15 of this application. We acknowledge that maintaining the privacy of interviewed sole developers, or members from a small developer team, may not be possible since their identities can be inferred. The interviewee will have to accept this risk if they choose to be interviewed. The risk of data breach will be minimized by not storing the interviewee names along with their answers, by gathering and storing all data on private password protected McMaster University systems, and by permanently deleting private password protected copies of data as detailed in Section 15 of this application.

Community Counselling or Support Services

14.7 Do you have a list of community counselling or other support services to give participants if they were to become distressed during participation in your research?

- ☐ Yes
- ☒ No

Deception and Partial Disclosure

14.9 Is there any deception or partial disclosure involved in this research?

- ☐ Yes
- ☒ No

Benefits of the Research Study

14.18 Note that many studies do not have DIRECT benefits to participants, and this is acceptable in most research contexts.

- a) Discuss the benefits that justify involvement of participants in this study in terms of any potential benefits to the scientific community, society in general, a specific population, and/or educational benefits (e.g. training students).
- b) If there are any direct benefits to the participants (e.g., participants receive skills training as part of the research methods), describe them.

Incentives to participate and reimbursements are NOT considered benefits of the research itself.

Receiving a copy of the study results is NOT considered a benefit for individual participants.

- a) The answers to these interview questions will help us analyze the current state of practice of software development in a domain, and provide insight into common or best practices for the benefit of future software development.
- b) There are not any direct benefits to the participants.

Experience with the Research

14.19 What is your experience with this kind of research? If this is student research, include the experience of your Faculty Supervisor.

Include information on the experience of **ALL** individual(s) who will have contact with the research participants or their data.

Describe experience with:

- (a) the proposed methods
- (b) the study population(s) and/or
- (c) the research topic.

- a) The Principal Investigator (Smith) has expertise in software engineering methods applied to Research software, as this has been the focus of his research since the year 2000. Smith has conducted multiple studies at McMaster that required ethics approval, such as one surveying the attitudes of scientific software developers and another, where student participants assessed the effectiveness of a computer game for teaching computing. Smith was the PI for a similar state of the practice assessment study conducted around 2020-2022. The Co-Investigator (Carette) also has experience with multiple empirical studies related to game design involving interviewing human participants. Carette was also a co-investigator the previous State of the Practice study. Both Smith and Carette are both extensively familiar with the ethical consideration of using human participants in research. The Student Principal Investigator (Ziyang) has academic and professional experience in the application of general software engineering methods.
- b) The study population is scientific computing software developers and the principal investigator and co-investigator are well familiar with this population as they have professional academic experience in this field and familiarity with similar software developers. The Student Principal Investigator (Ziyang) has familiarity with software developers through academic and professional experience.
- c) The Principal Investigator and Co-investigator have conducted research in this area for over twenty years combined. They have several publications related to this research area. The Student Principal Investigator (Ziyang) has been working on this research area for over a year.

Researchers are strongly encouraged to take the [TCPS2: Course on Research Ethics \(CORE\) tutorial](#) prior to conducting their study. Especially student researchers and researchers new to human participant research. If you have already completed the TCPS2: CORE tutorial, then you can include that training as part of your experience with human participant research in 14.19 above. Completion of the tutorial is not a requirement of the research ethics review process.

14.20 Risks to Researchers

Risks in research are not limited to participants. Researchers themselves could be exposed to risks due to study procedures (e.g., psychological risks from listening to participants recount their past traumatic events) or due to the overall goals of the project – e.g., a business/organization/government taking action against a researcher for critical research conducted about that institution. [At the end of Chapter 2 of the TCPS2](#), there is a short section highlighting that the REB may raise concerns about the safety of student researchers, and refer to the appropriate body within the institution. But ultimately it is not the REB's mandate to assess risks to researchers and require changes to mitigate those risks.

McMaster University has the following resources available to address risks to faculty, staff and students while conducting research, please follow-up as appropriate.

Research at a McMaster campus: For research taking place in McMaster labs, or on-campus in general, the [University Health and Safety](#) team can assist with assessing risks to researchers and planning risk mitigation.

Off-Campus Research: For research taking place off-campus, the University requires researchers to do the [RMM 801 process](#), which assesses risks and the risk management plan. For low risk situations, this is just handled as the department level, but for higher risk situations, University Health and Safety is involved.

Confidentiality and Data Security

The level of confidentiality and anonymity promised to participants can vary during different stages in the life cycle of the research (e.g., recruitment, consent, data collection, providing compensation or incentives, data preparation and analysis, data transfer/movement, data storage, dissemination of research findings and the final disposition of study documents [archiving or secure destruction, as applicable]). This section of the application covers the responsibility of researchers to describe their measures for meeting confidentiality obligations and for safeguarding participant information, per the [TCPS2, Articles 5.1, 5.2 and 5.3](#).

Confidentiality concerns the responsibility for the protection, privacy and security of information entrusted to researchers.

Anonymity concerns whether participant identities are known or not.

Please check the [MREB Data Storage and Security Tools](#) document in the Help - Templates menu above for best practices to secure electronic and hard copy versions of data and study documents.

Confidentiality and Security of Personal/Contact Information Collected for Administrative Purposes

15.1 Are you collecting personal information for administrative purposes during the recruitment, screening or consent phases of the study and/or to provide participants with incentives, reimbursements or study results?

Some examples of administrative documents with personal information include:

- Signed consent forms
- Email addresses for contacting participants
- Payment log
- Screening form
- Linking Document (has both participant identities and study ID)

☒ Yes

☐ No

Note that this series of questions is asking about personal information collected for administrative purposes, not research data. The confidentiality and security of research data is covered in the next section below.

15.2 What identifiable personal/contact information will be collected? (Check all that apply.)

- ☒ Name/signature
- ☒ Email addresses
- ☐ Phone number
- ☐ Mailing address
- ☐ Other

15.2.2 Describe the administrative purpose(s) for which this information is required (e.g. setting appointments, consent forms, consent log, distributing incentives, follow-up with participants for further research, sharing research results, etc.)

For our study, we will collect the first name, last name, and email address. The data will be used strictly for recruitment and correspondence. This administrative information may also be used to communicate study results as noted in the Letter of Information. The data will be available only to the Student Investigators, Principal Investigator, and Co-Investigator. The names/ email information will be stored in the password protected email accounts of the Student Investigators, Principal Investigator, and Co-Investigator. During the interview and throughout the entire study, only the above will be aware of the participants' identity.

Describe below the security procedures for the personal information collected for administrative purposes (e.g., consent forms, contact information). Keeping this information secure and confidential reduces potential risks for participants and is a requirement of privacy legislation (e.g., FIPPA). Refer to the [MREB Data Storage and Security Tools](#) for recommendations and requirements for data security (also found under Help - Templates in the above Help Menu).

15.3.1 Please describe:

- a) Who will have access to the personal information collected for administrative purposes (including people outside of the research team).
- b) If your answer to (a) includes people outside of the research team, then describe the information they have access to, why they need access, and what measures are in place so that participant confidentiality is maintained.

Note: Participants should consent to their personal information being shared outside the research team.

- a) Only the Student Investigators, Principal Investigator, and Co-Investigator.
- b) Not Applicable

15.3.2 How will the personal information be collected securely and how will it be kept secure during storage? If multiple members of the research team are collecting and storing the personal information, provide details on all collection and storage procedures, and how the personal information will be shared with other team members.

The names and emails will be stored on password protected email accounts of the Student Investigators, Principal Investigator, and Co-Investigator. They will also be shared using these password protected email accounts.

15.3.3 How will the personal information be transported securely from the field to the storage location, between McMaster and a home office, or other scenarios requiring the movement of the personal information? This could be hard copy documents (e.g., consent forms) or electronic (e.g., a laptop containing files with contact information).

Personal information will be stored on password protected email accounts. The personal information could potentially be transported on a password protected laptop if the aforementioned email accounts are signed into on the laptop. We will restrict the devices which will be used to access this email to personal devices. We will be aware of which endpoint devices may be used in accessing the secured email and we will ensure that these devices are password protected and clean of malware. We will not download emails onto devices. Email managers that access the emails will either have caching turned off or we will securely clear the cached copies of emails from the local endpoint.

- 15.4 Provide details on the retention of personal information for administrative purposes (e.g., consent forms, contact information). Please include:
- a) Length of time you plan on retaining the administrative documents containing personal/contact information. If some documents will be kept longer than others, please specify.
 - b) The reason why you need to retain the documents for the length of time stated in (a).
 - c) Details on how the documents will be destroyed at the end of the retention period.
 - d) If the procedures for longer term storage of personal information are different than what is described in 15.3, please provide the details.

Unless there is a reason to contact participants later, then contact information should be destroyed.

The retention period will span the length of the project so that all correspondence can be completed. It will be deleted upon the completion of the project. We will use a secure file deletion tool such as Shred (Linux/Unix) or Eraser (Windows).

Confidentiality and Security of Research Data

- 15.5 Are you collecting any research data that directly identifies participants (e.g., audio or video recording) or that could indirectly identify participants (e.g., a combination of demographic variables - date of birth, postal code, occupation, ethnicity, etc.)?

In this section "research data" refers to information collected from participants for analysis or to describe the sample.

- ☒ Yes
☐ No

- 15.8 Will there be a separate file with a unique code or pseudonym linking the participant name/contact information to the data (e.g., linking data between multiple sessions within a study, linking data in the current study to a future study)?

- ☐ Yes
☒ No

- 15.10 Select all (potentially) identifiable data that will be collected for this study and explain why each type of identifiable data is necessary to conduct the research. (Select all that apply)

- ☐ Full Postal Code
- ☐ Partial Postal Code
- ☐ Full Date of Birth
- ☐ Partial Date of Birth
- ☐ IP Address (ensure this function is turned off in online platforms if not collecting IP addresses)
- ☒ Audio Recording
- ☐ Video Recording
- ☐ Photographs
- ☒ Indirectly Identifying Variables (ethnicity, program of study, occupation, gender, etc.)
- ☐ Direct Quotes
- ☐ Other

15.10.6 Explain why it is necessary to audio record participants:

Audio recording will be required when conducting our interviews to capture all information accurately and also for the analysis of data. We will use Microsoft Teams' audio recording feature. The audio recording will be stored in our online password-protected data repository.

More details:

The audio recorded meeting will be automatically transcribed via the Student Investigator's, Principal Investigator's, or Co-Investigator's McMaster Microsoft Stream account. Microsoft Team and Microsoft Stream are linked, thus the meeting is automatically stored on Microsoft Stream. Transcription is done automatically in Microsoft Stream and the transcription file will be downloaded immediately after the interview is completed. The data on Microsoft Teams and Microsoft Stream are private data and not available to the public. The interviewer will log into the McMaster Microsoft stream account to download the original interview audio recording, and the transcribed file, then save on our online password-protected data repository for security and privacy. Only the Student Investigators, Principal Investigator, and Co-Investigator will have access to the stored data. After the original interview recording and the transcribed text have been securely stored on our online password-protected repository, we will permanently delete copies of the data from Microsoft Teams and Microsoft Stream.

15.10.9 List the indirectly identifying variables and explain why it is necessary to collect this information.

We will be collecting the participant's current position/titles and degrees. We will be collecting this as part of our qualitative data to better access the state of software development in the selected research software domains.

Describe below the data security procedures that will be used to keep the research data private and secure during data collection and analysis. Refer to the [MREB Data Storage and Security Tools](#) for recommendations and requirements for data security (go to Help - Templates in the above menu).

15.11.1 How will the research data be collected securely, to ensure participant privacy and confidentiality during the data collection phase of the study? If an online survey platform, video-conferencing platform, or similar will be used to collect the data, then provide information on how that service protects the data (this could include a link to the service's privacy policy).

The Student Investigators, Principal Investigator, and Co-Investigator will be collecting and storing research data. Interview data will be collected via Microsoft Teams App using audio recording. The interview will then be transcribed immediately after completion of the interview using McMaster's Microsoft Stream App. Microsoft Stream and Microsoft Teams are linked, hence there will be no need to manually transfer data. Transcription is automatically done in Microsoft Stream. When the transcription is completed, the interviewer will upload the audio interview recording and the corresponding transcribed data to our online password-protected data repository where only the Student Investigators, Principal Investigator, and Co-Investigator can access it. No other party will be able to access the data. The original audio recording on Microsoft Team and Microsoft Stream will be deleted permanently from the cloud servers and the transcribed data will also be deleted permanently from the Microsoft Stream cloud server.

15.11.2 How will research data be stored (e.g. digital files, hard copies, audio recordings, etc.)? Specify the physical and/or online storage location(s) and how data will be secured to minimize the risk of a data breach. If multiple members of the research team are storing the research data, provide details on all storage procedures and how the research data will be shared between team members.

Note: The TCPS2 advises that identifiable data obtained through research and kept on a computer that is connected to the Internet, should be encrypted. Generally, MREB requires researchers to follow this guidance for identifiable research data. In some cases, sensitive identifiable data, even if not on a device connected to the Internet, may need to be encrypted to protect participant privacy and mitigate social risk. See the application section of [TCPS2 Article 5.3](#).

Only the Student Investigators, Principal Investigator, and Co-Investigator will be collecting and storing data. All research data will be stored on our private password-protected data repository (Computing and Software Department Gitlab server) which can only be accessed by the Student Investigators, Principal Investigator, and Co-Investigator. We will download the interview audio recording and auto-transcribed text and store securely on our online password-protected data repository. Once the transcription has been validated for accuracy, the interviewer will delete the audio file. The interviewer will delete all remaining data permanently from the local machine or McMaster Microsoft Teams or Microsoft Stream account as applicable.

15.11.3 How will the research data be transported securely from the field to the storage location, between McMaster and a home office, or other scenarios requiring the movement of the research data (e.g. sending to a translator, downloading data from an online platform)? This could include both hard copy documents and electronic files.

All research data will be stored on our password-protected data repository (Computing and Software Department Gitlab server). We will download the interview audio recording and transcribed text and store securely on our online password-protected data repository. The interviewer will delete the data permanently from the local machine or McMaster Microsoft Teams or Microsoft Stream account as applicable. All data will be collected electronically, no hard copy required.

15.11.4 Please describe:

a) Who will have access to the research data (including people outside of the research team, e.g. translator, transcription service).

b) If your answer to (a) includes people outside of the research team, then describe the information they have access to, why they need access, and what measures are in place so that the data is kept secure.

Do not include sharing data to open source platforms or other repositories. This will be covered in a later question.

a) Only the Student Investigators, Principal Investigator, and Co-Investigator will have access to audio recordings and transcribed data.

b) Not Applicable

15.11.5 Will you be asking anyone who has access to research data, access to other study documents (e.g. contact lists), and/or knowledge of who participated in the study, to sign an oath of confidentiality?

Note: This is more common when someone outside the research team has access to confidential information (e.g. translator), or when some participants will know that other participants were in the study or what they said (e.g. in a focus group). But it may also include research team members in a study with high social risk for participants. The need for an [oath of confidentiality](#) depends on the study context, the sensitivity of the data, and the risk to participants if there was a loss of confidentiality.

☐ Yes

☒ No

Transfer of Study Records to Another Institution/Organization

15.12 Will any research data and/or study related documents be transferred to another institution or organization (e.g. to a research partner organization, to a separate research team at another university requesting use of the data, following OCAP principles, etc.)?

☐ Yes

☒ No

Dissemination of Findings and Final Disposition of Research Data

15.18 How will the data from study participants be reported in the dissemination of research results (e.g., aggregated data, identifiable descriptors, de-identified descriptors, etc.)?

• The software development process details that we gather will be linked to their software project, but we don't identify who provided the information. The disseminated data will thus use de-identified descriptors that link the data to a specific project but not to the study participant. While we will not identify participants, we acknowledge that maintaining the privacy of interviewed sole developers, or members from a small developer team may not be possible since their identities can be inferred. The interviewee will have to accept this risk if they choose to be interviewed.

15.19 Provide details on the retention of research data. Please include:

- a) Length of time you plan on retaining data. If some parts of the data will be kept longer than others, please specify.
- b) The reason why you need to retain the research data for the length of time stated in (a).
- c) If applicable, details on how the research data will be destroyed at the end of the retention period.

a) Data will be retained for 3 years after the study.
b) This should provide adequate time for analysis, the publication of results response to questions that may arise from published work.
c) The data will be removed i.e deleted permanently from our online password-protected repository thus it will be deleted permanently from the server.

15.20 Will you be retaining identifiable and/or coded research data long-term (i.e. beyond the initial data analysis phase)? Coded data refers to a de-identified data set that can be re-identified with a document linking participant ID numbers to names.

- ☐ Yes
☒ No

15.21 State at what point the data will be anonymized, or specify if the data was anonymous at the point of collection (e.g. an anonymous online survey).

Note: Data can only be considered anonymized once the information is irrevocably stripped of direct identifiers, a code is not kept to allow future re-linkage, and risk of re-identification of individuals from remaining indirect identifiers is low or very low, e.g., deletion of audio recordings or linking file.

Interview and project log data will be anonymized during dissemination in the report.

15.25 Do you have plans to have identifiable data professionally archived? (e.g. interview recordings kept in a library archive)

- ☐ Yes
☒ No

15.27 Will someone other than the Principal Investigator be retaining the study data long-term? In the case of student research, will someone other than the Student Investigator and/or Faculty Supervisor be retaining the data long-term?

- ☐ Yes
☒ No

15.29 Do you plan to post research data to a database accessible by other researchers and/or the general public, or to make data available to other researchers upon request? (e.g. some journals require sharing of data for verification purposes as a condition of publication)

- ☐ Yes
- ☒ No

Informed Consent

16.1 Describe the process the investigator(s) will use to obtain informed consent from participants with the capacity to provide consent. For participants lacking capacity to consent see the question below.

Please include;

- a) How participants will be informed of the study details prior to consenting (e.g. information letter via email, in-person by the researcher, reading letter online before a survey, etc.).
- b) How consent will be documented (e.g. signed consent form, verbal consent log, or clicking a button to indicate consent).
- c) Which member(s) of the research team will be handling the informed consent process, if applicable.
- d) Any unique consent details, for example if the study involves consent in multiple stages or an ongoing consent process.

- a) Letter of Information containing research study details will be sent to participants via email.
 - b) There will be a table that is filled out to track consent. This table will be filled out by the interviewer at the time of the interview for verbal consent in order to confirm consent to participate. This document will be stored on an online password protected McMaster University private repository. Copies will be temporarily stored on password protected computers, before being promptly deleted after being uploaded to the repository.
 - c) The Student Investigators will be handling the informed consent process.
 - d) The interviewer will verbally ask the interviewee for consent when they start the interview.

Alternative Consent Processes

16.2 Do any individuals lack the capacity, in the context of your study, to make an informed choice to participate in the research (e.g. children, people with cognitive impairments)?

- ☐ Yes
- ☒ No

16.4 Are you seeking an exception to the requirement that researchers seek consent from participants prior to the collection of data?

- ☐ Yes
- ☒ No

Informed Consent Documents

16.6 Select the document(s) that will be used in the consent process.

There are sample consent documents in the [Help - Templates](#) section in the menu above. You can sort the documents alphabetically by clicking on Document Name.

- ☒ Letter of Information / consent - Participants
- ☐ Online survey or study consent preamble
- ☒ Oral / telephone consent script
- ☒ Oral consent log
- ☐ Letter of Information / consent - Parent(s) or Guardian(s)
- ☐ Letter of Information / Assent form - Minors
- ☐ Letter of Information / consent - Substitute Decision Maker
- ☐ Not Applicable (e.g. secondary use of data, self study)
- ☐ Other

16.6.1 Upload your oral consent Log.

Upload in PDF format

The document file name should clearly indicate type of document, as the file name will appear on the clearance certificate.

If uploading a revised document,

- 1) use track changes so the revisions can be easily verified
- 2) delete the previous version of the document from this upload question.

Documents					
Type	Document Name	File Name	Version Date	Version	Size
Consent Forms	ConsentLog	ConsentLog.pdf	May/11/2026	1	95.0 KB

16.6.2 Upload your oral / telephone consent script.

Upload in PDF format

The document file name should clearly indicate type of document, as the file name will appear on the clearance certificate.

If uploading a revised document,

- 1) use track changes so the revisions can be easily verified
- 2) delete the previous version of the document from this upload question.

Documents					
Type	Document Name	File Name	Version Date	Version	Size
Consent Forms	Oral-Consent-Script	Oral-Consent-Script.pdf	May/11/2026	1	87.1 KB

16.6.3 Upload your Letter of Information / consent - Participants. Multiple data collection methods and/or participant groups may require multiple letters of information.

Upload in PDF format

The document file name should clearly indicate type of document, as the file name will appear on the clearance certificate.

If uploading a revised document,

- 1) use track changes so the revisions can be easily verified
- 2) delete the previous version of the document from this upload question.

Documents					
Type	Document Name	File Name	Version Date	Version	Size
Consent Forms	Letter-of-Information-Consent-Form2	Letter-of-Information-Consent-Form2.pdf	May/11/2026	1	68.3 KB

Providing Participants with Study Results

16.7 Will participants be able to learn about the study results (e.g., mailed/mailed brief summary of results in plain language; posting on website or other appropriate means for this population)?

- ☒ Yes
☐ No

16.9 Explain how participants will learn about study results:

The participants will learn about the study results through the research paper. The paper should be complete by December 2026. If desired, a copy will be sent to each requester.

Participant Withdrawal

16.10 Will participants have the right to withdraw from the study during their data collection (e.g. during an interview, during a lab session)?

- ☒ Yes
☐ No

16.10.1 Describe:

- a) how the participants will be informed that they can withdraw during the data collection process, and
b) the procedures which will be followed to allow participants to exercise this right.

a) The Letter of information/consent explicitly states their right to withdraw.
b) All that would be required to withdraw would be an e-mail to a Student investigator or Principle Investigator. The withdrawal will be acknowledged via email in response to the withdrawal, or verbally at the point of contact.

16.10.2 For a participant who withdraws during the data collection process, describe:

- a) what will be done with any data collected up to the point of withdrawal, and
b) consequences withdrawal might have on the participant, including any effect that withdrawal may have on the participant(s) incentive/reimbursement or continuation of services (if applicable).

a) If a participant withdraws, all of their data will be deleted, if they so wish.
b) There is no compensation for participation.

16.11 Will participants have the right to withdraw their data from the study after their data has been submitted/collected?

- ☒ Yes
☐ No
☐ Both Yes and No (e.g. due to multiple data collection methods)

16.11.1 Describe:

- a) at what point/date after data collection it will no longer be possible to withdraw data from the study,
- b) how participants can request data be removed before this point/date, and
- c) if there will be any consequences for participants withdrawing their data.

Be sure to include these details in the Letter of Information.

- a) Participants are able to withdraw data from the study at any time before the data has been analyzed into the study.
- b) Participants can request data to be removed by sending an email to a Student investigator or Principal Investigator. They can also withdraw during the interview if they so wish.
- c) There will be no consequences for withdrawing.

Incidental Findings and Third Party Disclosure

16.13 Is there a potential of material incidental findings resulting from your research? See the info button for further details (most studies reviewed by MREB will not have incidental findings).

- ☐ Yes
- ☒ No

16.15 Is there a reasonable possibility the researcher will obtain information from participants that will require the researcher to break confidentiality and report details to a third party? This could be a legal or ethical requirement (e.g. suspected child abuse, imminent self-harm or harm to others). See the info button for further details.

- ☐ Yes
- ☒ No

Additional Information

18.1 Do you have any additional information or documents relevant to this project that you wish to provide to the Research Ethics Board?

- ☐ Yes
- ☒ No

Posting of Approved Protocols on the Research Ethics Website

18.4 Public posting your research title

It is the policy of MREB to post a list of cleared protocols on the Research Ethics website. Posted information usually includes: title, name of principal investigators, principal investigator department, type of project (i.e. Faculty; PhD; Masters, Undergraduate etc.)

Do you request that the title be deleted from the posted information?

- ☐ Yes
- ☒ No

Principal Investigator Assurance(s)

Principal Investigator Assurances

I understand that the following all constitute violations of the McMaster University's Research Integrity Policy:

- failure to obtain research ethics clearance;
- carrying out research in a manner that was not cleared by one of the university's REBs ([see TCPS2, Art. 6.11](#));
- failure to submit an **Amendment** to obtain ethics clearance prior to implementing substantive changes to a cleared study ([see TCPS2, Art. 6.16](#));
- failure to submit an **Annual Project Status form** in advance of the yearly anniversary of the original ethics clearance date; ([see TCPS2, Art. 6.14](#)).

Additionally, researchers are required to report **Adverse Events** (i.e., an unintended negative consequence or result affecting participants) to the MREB ethics office and the MREB Chair as soon as possible, and no more than 3 days after the event occurs ([see TCPS2, Art. 6.15](#)). A privacy breach affecting participant information should also be reported to the MREB ethics office and the MREB Chair as soon as possible. The Reportable Events form is used to document adverse events, privacy breaches, protocol deviations and participant complaints.

I confirm that I have read the [McMaster University Research Integrity Policy](#), and I agree to comply with this and other university policies, guidelines and the Tri-Council Policy Statement (TCPS2) and the guidelines of my profession or discipline regarding the ethical conduct of research involving humans.

20.2 Signature of Principal Investigator